Mermin Wagner thm& SnTe TCI

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Mermin Wagner theorem

General

Continuous symmetries cannot be spontaneously broken at finite temperature in systems with sufficiently short-range interactions in dimensions $d \leq 2$

Applied in magnetism,

Magnetism

At finite temperature, isotropic quantum Heisenberg model with short range exchange interaction on $d \leq 2$ lattices have no FM or AFM order.

Conditions

- Finite temperature: $T > 0$
- Isotropic Heisenberg model:

$$H = \sum_{ij} J s_i^x s_j^x + J s_i^y s_j^y + J s_i^z s_j^z, \; J_x = J_y = J_z = J$$

  no structure an-isotropy $\sum_i (s_i^z)^2$, no SOC
- Short range exchange interaction:

$$J_{ij} \sim \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|^\alpha}, \; \alpha > 1, \quad \frac{1}{N} \sum_{ij} |J_{ij}||\boldsymbol{r}_i - \boldsymbol{r}_j|^2 < \infty$$

Bogouliubov inequality

We first prove the Bogouliubov inequality:

$$\frac{\beta}{2}\langle\{A, A^\dagger\}\rangle\langle\left[C^\dagger, [H, C]\right]\rangle \geq \left|\langle[C^\dagger, A^\dagger]\rangle\right|^2$$

Consider an inner product defined as:

$$(A, B) = \sum_{n\neq m}\langle n|A^\dagger|m\rangle\langle m|B|n\rangle\frac{p_m - p_n}{E_n - E_m}, \quad p_n = \frac{e^{-\beta E_n}}{Z}$$

Bogouliubov inequality

The inner product defined above has following properties:

$1)(A,B)=(B,A)^* \quad 2)(\alpha A+\beta B,C)=\alpha(A,C)+\beta(B,C) \quad 3)(A,A)\geq 0$

So it satisfies Schwartz inequality:

$$(A,A)(B,B)\geq |(A,B)|^2$$

Now set $B=[C^\dagger,H]$, easy to show that:

$$(B,B)=\langle\left[C^\dagger,[C,H]\right]\rangle$$

Bogouliubov inequality

Also

$$
\begin{aligned}
(A,B) &= \sum_{n\neq m} \langle n|A^\dagger|m\rangle\langle m|[C^\dagger,H]|n\rangle \\
&= \sum_{n\neq m} \langle n|A^\dagger|m\rangle\langle m|C^\dagger|n\rangle(p_m-p_n) = \langle[A^\dagger,C^\dagger]\rangle
\end{aligned}
$$

and since when $E_n > E_m$:

$$
\begin{aligned}
\frac{p_m-p_n}{E_n-E_m} &= \frac{1}{Z}\frac{e^{-\beta E_m}-e^{-\beta E_n}}{E_n-E_m} = \frac{1}{Z}\frac{e^{-\beta E_m}+e^{-\beta E_n}}{E_n-E_m}\cdot\frac{e^{-\beta E_m}-e^{-\beta E_n}}{e^{-\beta E_m}+e^{-\beta E_n}} \\
&= \frac{1}{Z}\frac{e^{-\beta E_m}+e^{-\beta E_n}}{E_n-E_m}\cdot\tanh\frac{\beta}{2}(E_n-E_m) \\
&< \frac{1}{Z}\frac{e^{-\beta E_m}+e^{-\beta E_n}}{E_n-E_m}\cdot\frac{\beta}{2}(E_n-E_m) = \frac{\beta}{2}(p_m+p_n)
\end{aligned}
$$

Bogouliubov inequality

$$
\begin{aligned}
(A, A) &= \sum_{n \neq m} \langle n|A^\dagger|m\rangle\langle m|A|n\rangle \frac{p_m - p_n}{E_n - E_m} \\
&\leq \sum_{n \neq m} \langle n|A^\dagger|m\rangle\langle m|A|n\rangle \cdot \frac{\beta}{2}(p_m + p_n) \\
&\leq \frac{\beta}{2} \sum_{nm} \langle n|A^\dagger|m\rangle\langle m|A|n\rangle (p_m + p_n) \\
&= \frac{\beta}{2}\langle AA^\dagger + A^\dagger A\rangle = \frac{\beta}{2}\langle\{A, A^\dagger\}\rangle
\end{aligned}
$$

Plug these into Schwartz inequality, we get Bogouliubov inequality:

$$
\frac{\beta}{2}\langle\{A, A^\dagger\}\rangle\langle\Big[C^\dagger, [H, C]\Big]\rangle \geq \Big|\langle[C^\dagger, A^\dagger]\rangle\Big|^2
$$

Mermin Wagner thm

Spin S Heisenberg model: $H = \sum_{ij} J_{ij} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + h \sum_i S_i^z$
Set $C = S_k^x = \sum_j e^{ik\cdot r_i} S_i^x$, $A = S_{-k-q}^y$, $m_q = \frac{1}{N}\langle S_q^z \rangle$, we get

$$\frac{1}{2}\langle \{A, A^\dagger\} \rangle = \langle S_{k+q}^y S_{-k-q}^y \rangle, \quad \langle [C^\dagger, A^\dagger] \rangle = iNm_q$$

Put into Bogouliubov inequality and set

$$S^{yy}(k+q) = \frac{1}{N}\langle S_{k+q}^y S_{-k-q}^y \rangle, \quad F(k) = \frac{1}{N}\langle \left[S_{-k}^x, [H, S_k^x]\right] \rangle$$

we get

$$S^{yy}(k+q)F(k) \geq Tm_q^2$$

Mermin Wagner thm

Set $\bar{J} = \frac{1}{N}\sum_{ij}|J_{ij}||\boldsymbol{r}_i - \boldsymbol{r}_j|^2$, it can be shown that

$$F(k) \leq hm_q + S(S+1)\bar{J}|k|^2$$

So

$$S^{yy}(k+q) \geq \frac{Tm_q^2}{hm_q + S(S+1)\bar{J}|k|^2}$$

Since $\frac{1}{N}\sum_k S^{yy}(k+q) \leq S(S+1)$, we get

$$\boxed{S(S+1) \geq \frac{1}{N}\sum_k \frac{Tm_q^2}{hm_q + S(S+1)\bar{J}|k|^2}}$$

Mermin Wagner thm

$$\boxed{\frac{S^2(S+1)^2\bar{J}}{T} \geq m_q^2 \cdot I(a), \quad I(a) = \frac{1}{N}\sum_k \frac{1}{k^2+a^2}, \quad a = \frac{hm_q}{S(S+1)\bar{J}}}$$

At thermodynamic limit, $I(a) \to \frac{1}{(2\pi)^d}\int_0^{k_0} \frac{k^{d-1}}{k^2+a^2}dk$, and turn off magnetic field $h = 0, a \to 0$

$$d = 1, I(a) \sim \frac{1}{a}\arctan\left(\frac{k_0}{a}\right) \to \frac{1}{a}\frac{\pi}{2} \to \infty$$

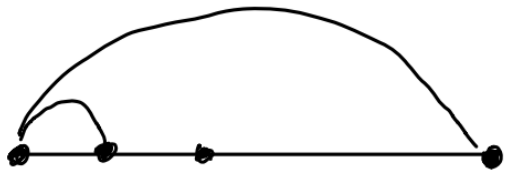
$$d = 2, I(a) \sim \ln\left(1 + \frac{k_0^2}{a^2}\right) \to \infty$$

$$d = 3, I(a) \sim k_0 - a\arctan\left(\frac{k_0}{a}\right) \to k_0$$

Contradicted with the inequality when $d \leq 2$, proved.

Finite size effects

What if the system is of finite size? i.e., $I(a)$ is a discrete sum. Then there is a lower bound of possible momentum: $k_{min} = \frac{\pi}{N^{1/d}}$, because there is an upper bound of spin wave length.



$$\text{When } d = 1, I(a) = \frac{1}{N} \sum_{n \neq 0, n=-N}^{N} \frac{1}{\frac{n^2\pi^2}{N^2} + a^2}$$

$$I(0) = \frac{2N}{\pi^2} \sum_{n=1}^{N} \frac{1}{n^2} \approx \frac{2N}{\pi^2} \cdot \frac{\pi^2}{6} = \frac{N}{3}$$

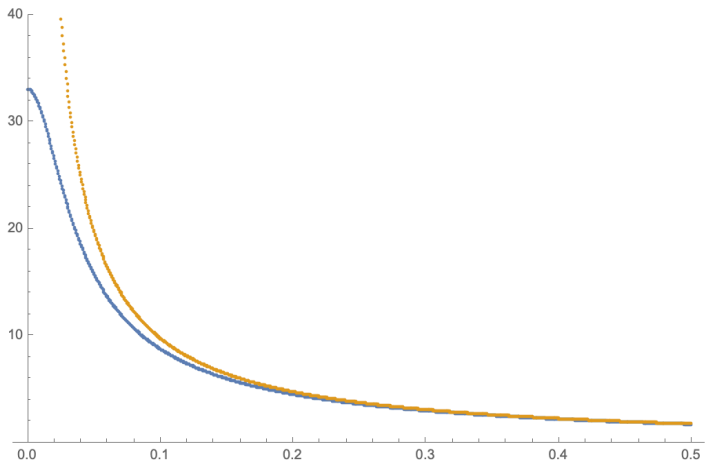
So there is an upper bound of magnetization

$$m_{max} = S(S+1)\sqrt{\frac{3\bar{J}}{NT}}$$

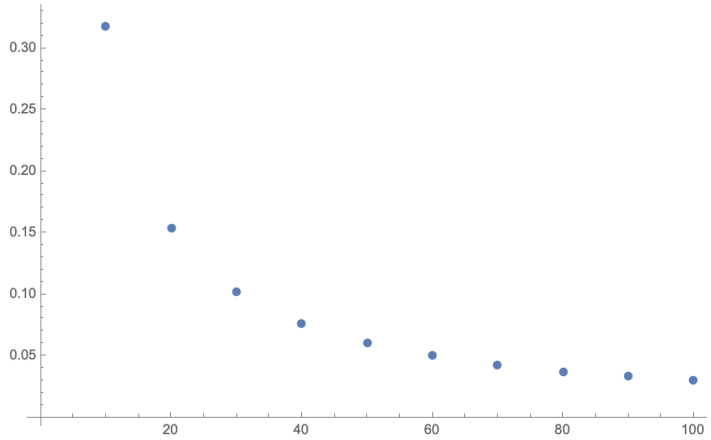
For example, choose $\bar{J} = 10meV, T = 1K, N = 100$, then $m_{max} = 1.4$

Finite size effects in 1d

$I(a)$ vs a plot, $N = 100$, discrete sum and integral values



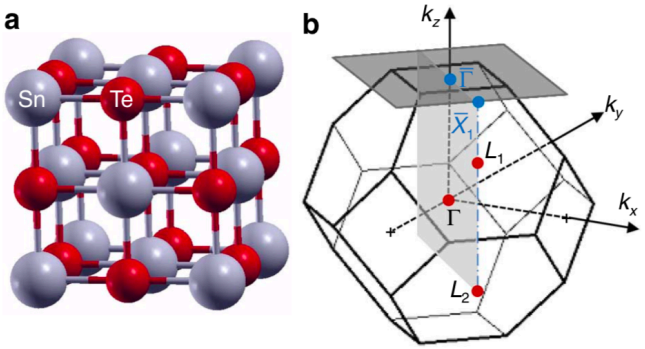
$m_{max} \sim 1/I(0)$ vs N plot, $1/N$ decay



Reason for no magnetic order in 1d and 2d

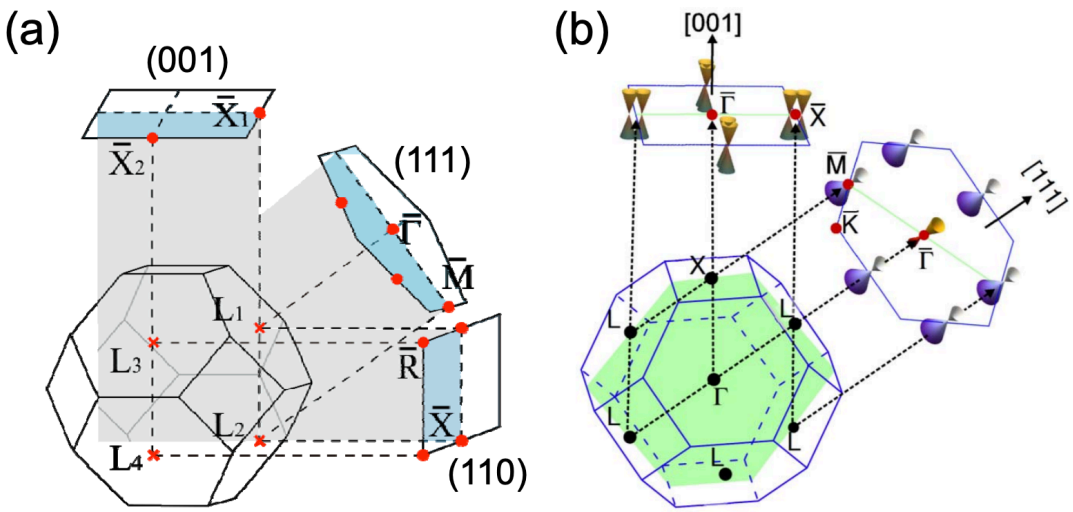
- Due to less number of neighboring sites, the fluctuations in 1d and 2d have greater destruction of magnetization.
- Usually a spin $1/2$ chain with length L will has excitation gap $\sim 1/L^{\alpha}$, so that in thermodynamic limit it becomes gapless. A small fluctuation can cause an excitation that propagates through long range.

SnTe

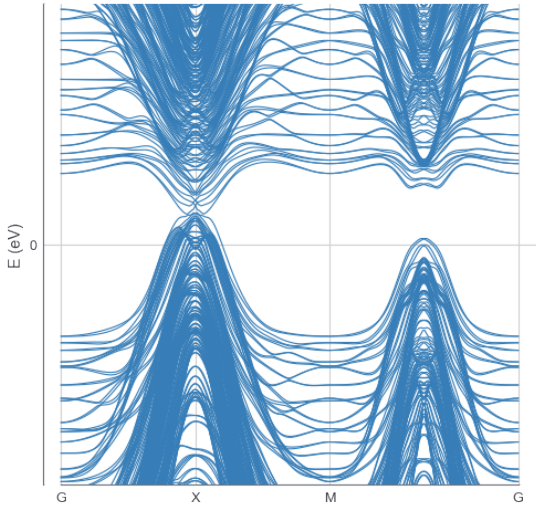


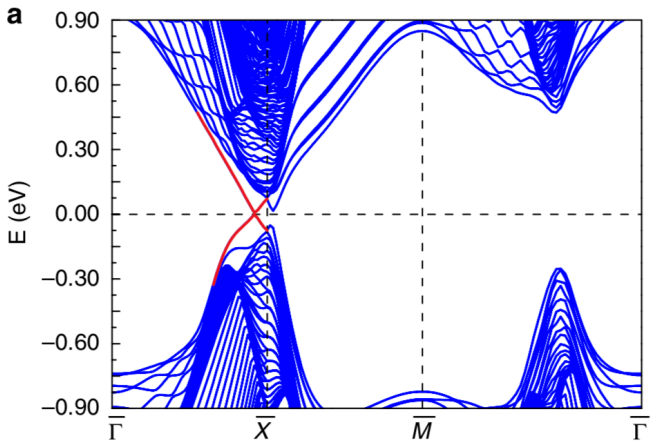
- SnTe is a topological crystalline insulator with mirror symmetry w.r.t. $\{110\}$ and equivalent planes;
- Even number of Dirac cones on $\{001\}$,$\{110\}$,$\{111\}$ and equivalent planes which are invariant under mirror reflections in $\{110\}$ mirror planes;
- Although the total Chern number summed up to be 0, the mirror Chern number of each invariant plane is $|n_M| = 2$

SnTe



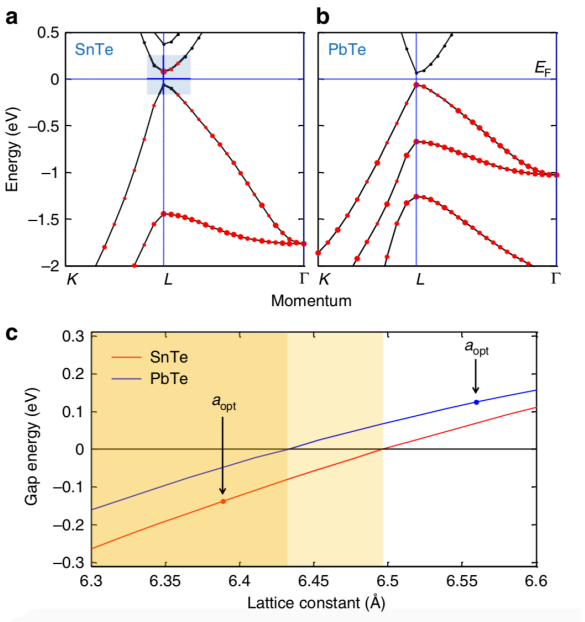
One Dirac cone along $\bar{\Gamma} - \bar{X}$, 10 layers and hundreds of layers:





SnTe and PbTe

- PbTe is a band insulator, it can be smoothly connected to vacuum
- SnTe is a TCI, it can not be smoothly connected to a band insulator without closing the gap
- Changing from PbTe to SnTe, there is a gap closing and re-opening process, and a band inversion



- For $Sn_xPb_{(1-x)}Te$, previous numerical results showed that critical x_c is 0.38. The surface states will disappear when $0 \leq x \leq x_c$.